ECOBUILD AI

AI-Powered Sustainable Construction Intelligence

Audit Date: November 09, 2026
Standard: ISO 14040 / GHG Protocol / ASHRAE

Project Details	Sustainability Rating
Project Name: high heights **Building Type:** Residential **Location:** pune **Gross Built-up Area:** 1200.0 m² **Floors / Occupants:** 3 floors / 24 people **Planned Construction:** 2026	**86.2/100** **Classification:** Gold • Energy Performance: **85.0/100** • Carbon Footprint: **100.0/100** • Water Conservation: **90.8/100** • Materials Circularity: **55.2/100**

1. Environmental Assessment Summary

Category	Key Metric	Assessed Value	Standard Benchmark	Rating / Status
Energy	Energy Use Intensity (EUI)	79.17 kWh/m²/yr	110.0 kWh/m²/yr	Class A
Renewables	Annual Solar Generation	0 kWh	Pot: 27,840 kWh	Undeployed
Water	Per Capita Consumption	133.33 LPCD	135.0 LPCD	Optimized
Rainwater	Rooftop Catchment Yield	231,200 L/yr	850 mm/yr Precipitation	High Capture Potential
Operational Carbon	Scope 2 Grid Emissions	68.02 tCO2e/yr	Grid: 0.716 kg/kWh	Annual Operational
Embodied Carbon	Cradle-to-Gate Materials	142.56 tCO2e	Amortized over 50 yrs	Concrete: 86.41 t | Steel: 51.09 t
Carbon Intensity	Total Annual Footprint	59.06 kg CO2e/m²	Target: < 90 kg/m²	Compliant

2. Prioritized AI Sustainability Recommendations

No active remediation measures identified for current design baseline.

3. Calculation Methodology & Reference Standards

Governing Standards & Emission Factors:

• **Operational Carbon:** Evaluated under GHG Protocol Scope 2 location-based accounting utilizing grid emission intensity factor of 0.716 kg CO2e/kWh.

• **Embodied Carbon:** Cradle-to-gate inventory derived from the University of Bath Inventory of Carbon and Energy (ICE v3.0). Standard ready-mix concrete rated at 240 kg CO2e/m³, structural steel at 1.85 kg CO2e/kg (reduced proportionally for recycled EAF scrap content).

• **Water & Rainwater:** Calculations follow National Building Code (NBC) standard baseline of 135 LPCD for residential dwellings and 45 LPCD for commercial workplaces. Rainwater harvesting assumes local precipitation coefficient with 85% impervious runoff and 80% secondary filtration efficiency.

• **Limitations:** This report presents an engineering estimation model for early-stage design optimization. Actual commissioned performance may vary based on weather variations, microclimate conditions, and tenant behavioral patterns.